REPLACE AND MODERNIZE LEGACY VPN ARCHITECTURES WITH A CLOUD-NATIVE ZSCALER ZERO TRUST SOLUTION TO REDUCE ATTACK SURFACE AND LATERAL MOVEMENT.

Project Details

Title:Replace and modernize legacy VPN architectures with a cloud-native Zscaler Zero Trust solution to reduce attack surface and lateral movement.

In today's digital environment, organizations require secure remote access for employees and users to connect to internal applications and resources. Traditionally, Virtual Private Networks (VPNs) have been used to provide this access by creating encrypted connections between users and the organization's network. However, many legacy VPN architectures follow a perimeter-based security model where users who successfully authenticate are given broad access to the internal network. This approach increases security risks, as attackers who gain access can move laterally within the network and potentially reach sensitive systems.

With the growth of cloud services and remote work, traditional VPN solutions are becoming less effective in protecting modern IT environments. To address these challenges, organizations are adopting the **Zero Trust security model**, which follows the principle of **"never trust, always verify."** Instead of providing full network access, Zero Trust continuously verifies user identity and allows access only to specific applications.

This project focuses on replacing traditional VPN infrastructure with a **cloud-native Zscaler Zero Trust solution**. By providing secure, identity-based access to applications rather than the entire network, the system reduces the attack surface and prevents unauthorized lateral movement. This approach improves overall security while supporting modern remote and cloud-based work environments.

Abstract:

In modern organizations, secure remote access to internal applications and data has become an important requirement due to the growth of cloud computing and remote work environments. Traditionally, Virtual Private Networks (VPNs) have been used to provide remote connectivity. However, many legacy VPN architectures follow a perimeter-based security approach that allows users broad access to internal networks after authentication. This increases security risks, as attackers who gain access through compromised credentials or devices can move laterally within the network and potentially access sensitive resources.

To address these challenges, organizations are adopting the **Zero Trust security model**, which is based on the principle of "never trust, always verify." Instead of trusting users simply because they are inside the network, Zero Trust continuously verifies user identity, device security, and access permissions before granting access to applications.

This project focuses on replacing traditional VPN infrastructure with a **cloud-native Zero Trust solution using Zscaler**. The Zscaler platform provides secure and direct access to applications without exposing the internal network to the internet. By connecting users only to the applications they are authorized to use, the system significantly reduces the attack surface and prevents unauthorized lateral movement.

The proposed approach enhances network security, improves access control, and supports modern cloud-based environments. Overall, implementing a Zscaler Zero Trust architecture helps organizations strengthen cybersecurity, protect sensitive data, and provide secure remote access in a scalable and efficient manner.

Use Case Scenarios

Scenario 1: Remote Employee Access to Internal Application

An employee working from home needs access to the organization's internal HR application. Instead of connecting through a traditional VPN, the user connects using the Zscaler Zero Trust solution. The system verifies the user's identity and device security before granting access. The user is only allowed to connect to the HR application and cannot access other internal resources.

Output:

• User identity verified (Employee_User_Group)
• Secure authentication through identity provider
• Connection restricted to HR Application only
• Internal network remains hidden
• No lateral movement to other systems possible

Scenario 2: Vendor Access to Internal System

A third-party vendor requires temporary access to a specific internal system to perform maintenance tasks. Using the Zscaler Zero Trust model, the vendor is provided limited access only to the required application rather than the entire network.

Output:

• User identity verified (Vendor_User_Group)
• Time-limited access granted
• Connection restricted to maintenance system
• Network segmentation maintained
• Unauthorized resources remain inaccessible

Scenario 3: Student Access to Research Server

A research student needs to access a university server to run data analysis scripts. Through Zscaler Zero Trust, the student can securely connect to the research server without exposing the entire university network.

Output:

• User identity verified (Research_Student_Group)
• Secure micro-tunnel connection established
• Access limited to research server only
• Other network services blocked
• Secure encrypted communication maintained

Scenario 4: IT Administrator Access to Servers

An IT administrator needs to manage internal servers for maintenance and monitoring. With Zero Trust architecture, the administrator can access specific servers securely without exposing management ports to the internet.

Output:

• User identity verified (IT_Admin_Group)
• Privileged access policy enforced
• Secure connection to specific server
• Activity monitoring and logging enabled
• No access to unrelated systems

Scenario 5: Guest User Access to Library Resources

A visiting faculty member requires temporary access to digital library resources hosted on the internal server. The Zero Trust solution allows access only to the library application without connecting the user to the entire network.

Output:

• User identity verified (Guest_Faculty_Group)
• Temporary session access provided
• Connection restricted to Library Server
• Network segmentation maintained
• Sensitive systems remain protected

Stage 1

Title of the project :

Replace and modernize legacy VPN architectures with a cloud-native Zscaler Zero Trust solution

Overview :

Traditional Virtual Private Network (VPN) architectures were designed to provide secure remote access to internal corporate networks. However, these legacy systems often grant users broad network access once connected, which increases the risk of unauthorized access, lateral movement, and potential data breaches. With the rapid adoption of cloud computing, remote work, and mobile devices, traditional VPN solutions are no longer sufficient to secure modern IT environments.

To overcome these limitations, organizations are adopting the **Zero Trust Architecture**, which follows the principle of "never trust, always verify." In this model, every user, device, and application must be authenticated and authorized before access is granted. Instead of allowing full network access like traditional VPNs, Zero Trust provides secure and controlled access only to specific applications.

A cloud-native platform such as the **Zscaler Zero Trust Exchange**, developed by **Zscaler**, helps organizations replace legacy VPN architectures with a more secure and scalable solution. It connects users directly to applications without exposing the internal network and continuously verifies user identity, device posture, and security policies. This approach reduces the attack surface and improves overall security.

By modernizing legacy VPN infrastructure with a Zero Trust solution, organizations can enhance data protection, support secure remote access, and improve user experience. It also simplifies network management and provides better scalability for cloud-based applications and distributed work environments.

List of teammates:

S.NO	NAME	COLLEGE	CONTACT
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List of Vulnerability Table:

S.NO	CATEGORY NAME	VULNERABILITY NAME	CWE/CVE - NO
1	Broken Access Control	Missing Authorization	CWE-862
2	Security Misconfiguration	Use of Hard-coded, Security-relevant Constants	CWE-547
3	Software Supply Chain Failures	Reliance on Component That is Not Updateable	CWE- 1329

4	Cryptographic Failures	Key Exchange without Entity Authentication	CWE-322
5	Injection	Improper Neutralization of HTTP Headers for Scripting Syntax	CWE-644
6	Insecure Design	Trust Boundary Violation	CWE-501
7	Authentication Failures	Authentication Bypass by Alternate Name	CWE-289
8	Software or Data Integrity Failures	Insufficient Verification of Data Authenticity	CWE-345
9	Security Logging & Alerting Failures	Information Loss or Omission	CWE-221
10	Mishandling of Exceptional Conditions	NULL Pointer Dereference	CWE-476

REPORT:

1.Vulnerability name: Missing Authorization

CWE/CVE: 862

OWASP/SANS category: OWASP Top 10 – Broken Access Control

Description:

This vulnerability, known as **CWE-862: Missing Authorization**, arises when a web application fails to enforce proper access restrictions on its users. In such cases, the system does not verify whether the current user has the necessary permissions before granting access to a resource or allowing an action. As a result, attackers can exploit this weakness by manipulating URLs, parameters, or API requests to bypass restrictions and gain unauthorized access to sensitive information.

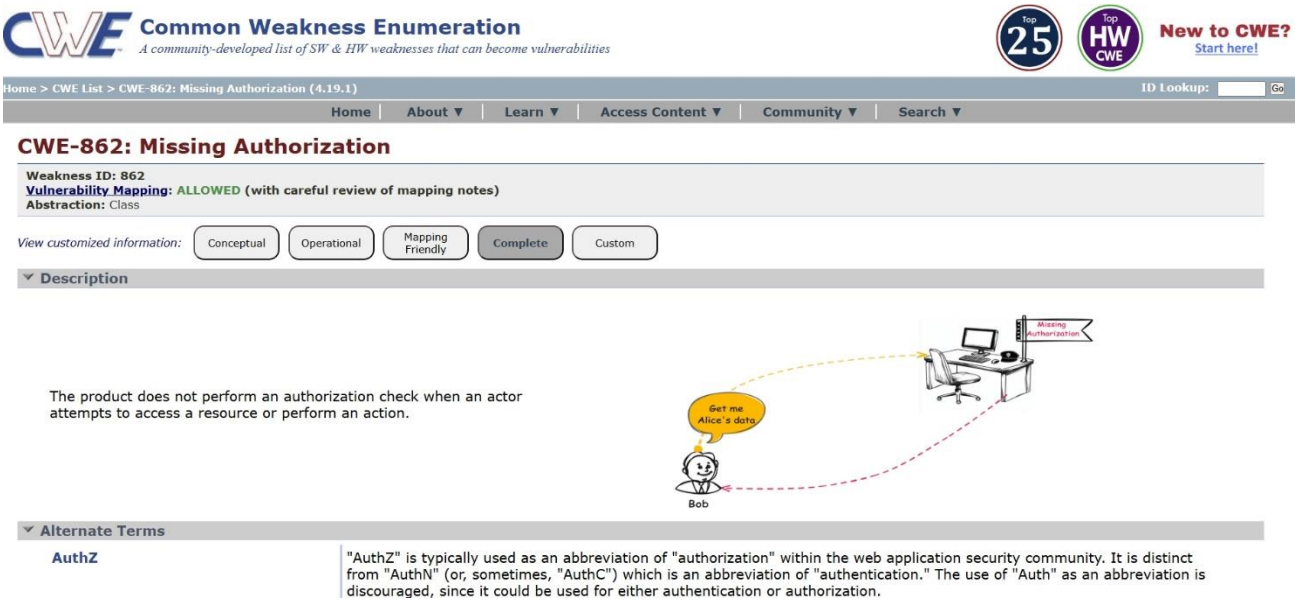
This may include personal details, account records, or confidential system data belonging to other users. The issue highlights the critical difference between authentication (verifying identity) and authorization (verifying privileges). While authentication ensures that a user is legitimate, authorization ensures that the user is only allowed to perform actions or view data within their assigned role. When authorization checks are missing, even authenticated users can escalate privileges or view data they should not, leading to serious breaches of privacy and security.

Vulnerability path:

http://testfire.net/

Business Impact:

When authorization checks are missing, attackers can exploit the weakness to gain access to sensitive customer or organizational data. This exposure of confidential information creates serious security and privacy risks, as attackers may steal personal records, manipulate account details, or even take unauthorized control of user accounts. Such incidents can lead to direct financial losses through fraud, as well as legal penalties for failing to comply with data protection regulations. Beyond the immediate damage, the organization's reputation suffers because customers and partners lose trust in its ability to safeguard information. In industries where compliance is critical, such as healthcare or finance, a single breach can also result in regulatory violations and long-term business setbacks.



Recommendation / Mitigation:

- Implement **strong access control mechanisms** so that users can only access resources they are authorized to use.
- Validate **all user requests on the server side**, ensuring authorization checks are performed before granting access.
- Avoid exposing **sensitive information** such as user IDs, account numbers, or internal system data in URLs or request parameters.
- Use **secure authentication and proper session management** techniques to protect user sessions and prevent hijacking.
- Apply the **principle of least privilege**, granting only the minimum permissions necessary for users and applications
- Conduct **regular security testing** such as vulnerability scanning, penetration testing, and code reviews to identify and fix weaknesses.
- Implement **logging and monitoring mechanisms** to detect suspicious activities and unauthorized access attempts.

2. Vulnerability Name: Use of Hard-coded, Security-relevant Constants
CWE/CVE: CWE-547

OWASP/SANS Category: OWASP Top 10- Security Misconfiguration

Description:

This vulnerability occurs when developers use hard-coded constants for security-critical values instead of symbolic names or configurable parameters. By embedding fixed values directly into the code, the system becomes difficult to maintain and prone to errors during updates or policy changes. If all occurrences of the constant are not identified and modified consistently, incorrect security decisions may result, weakening the application's defenses. Furthermore, once the code is released, changing these values requires code modifications that may be impractical or impossible in production environments. In cases where the source code is disclosed, attackers can easily discover these hard-coded values and exploit them to bypass security controls. The use of hard-coded constants not only reduces flexibility and maintainability but also introduces risks of information leakage and poor security practices, making the system more vulnerable to attacks..

Vulnerability Path:

http://example.com/debug

Business Impact:

When developers rely on hard-coded security-relevant constants, the organization faces significant risks in both security and maintainability. Any change in security policy or configuration requires code modifications, which may be difficult or impossible once the system is deployed. If all occurrences of the constant are not updated consistently, incorrect policy decisions may be made, weakening the application's defenses. In addition, if the source code is ever disclosed, attackers can easily identify these hard-coded values and exploit them to bypass security controls. This vulnerability can lead to unexpected behavior, reduced flexibility, and the introduction of new weaknesses during code maintenance. For businesses, the impact includes degraded software quality, increased operational costs, and higher risk of security breaches. Over time, reliance on hard-coded constants can damage the organization's reputation, reduce customer trust, and create compliance challenges if sensitive values are exposed or mismanaged.

Common Weakness Enumeration

A community-developed list of SW & HW weaknesses that can become vulnerabilities

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CWE-547: Use of Hard-coded, Security-relevant Constants

Weakness ID: 547
Vulnerability Mapping: ALLOWED
Abstraction: Base

View customized information: Conceptual | Operational | Mapping Friendly | Complete | Custom

Description

The product uses hard-coded constants instead of symbolic names for security-critical values, which increases the likelihood of mistakes during code maintenance or security policy change.

Extended Description

If the developer does not find all occurrences of the hard-coded constants, an incorrect policy decision may be made if one of the constants is not changed. Making changes to these values will require code changes that may be difficult or impossible once the system is released to the field. In addition, these hard-coded values may become available to attackers if the code is ever disclosed.

Common Consequences

Impact	Details
Varies by Context; Quality Degradation; Reduce Maintainability	**Scope: Other** The existence of hardcoded constants could cause unexpected behavior and the introduction of weaknesses during code maintenance or when making changes to the code if all occurrences are not modified. The use of hardcoded constants is an indication of poor quality.

Potential Mitigations

Phase(s)	Mitigation

Recommendation / Mitigation:

- Replace **hard-coded constants** with symbolic names, configuration files, or environment variables that can be updated without modifying source code.
- Adopt **centralized configuration management** so that critical values are maintained consistently across the application.
- Store sensitive constants securely, for example in **encrypted configuration files** or secure key management systems.
- Perform **regular code reviews and security testing** to identify and eliminate hard-coded values that weaken flexibility and security.
- Ensure **policy changes** can be applied without requiring direct code modifications, improving maintainability and reducing risk.
- Train developers to follow **secure coding practices**, avoiding shortcuts like embedding fixed values directly in the code.

3. **Vulnerability Name:** Reliance on Component That is Not Updateable

CWE/CVE: 1329

OWASP/SANS Category: OWASP Top 10 – Software Supply Chain Failures

Description:

This vulnerability occurs when an application depends on third-party software components, libraries, or frameworks that cannot be updated or patched easily. Modern applications often rely on external packages to provide additional features and functionality. However, if these components are outdated, unsupported, or not designed to be updated, they may contain known security vulnerabilities that remain unpatched.

Attackers can exploit these weaknesses because the organization cannot apply security fixes or upgrade the affected component. As a result, the system becomes exposed to potential attacks such as unauthorized access, code execution, or data compromise. Reliance on components that cannot be updated increases the overall security risk and makes it difficult to maintain a secure software environment.

Vulnerability Path:

http://example.com/app/plugins/old-library

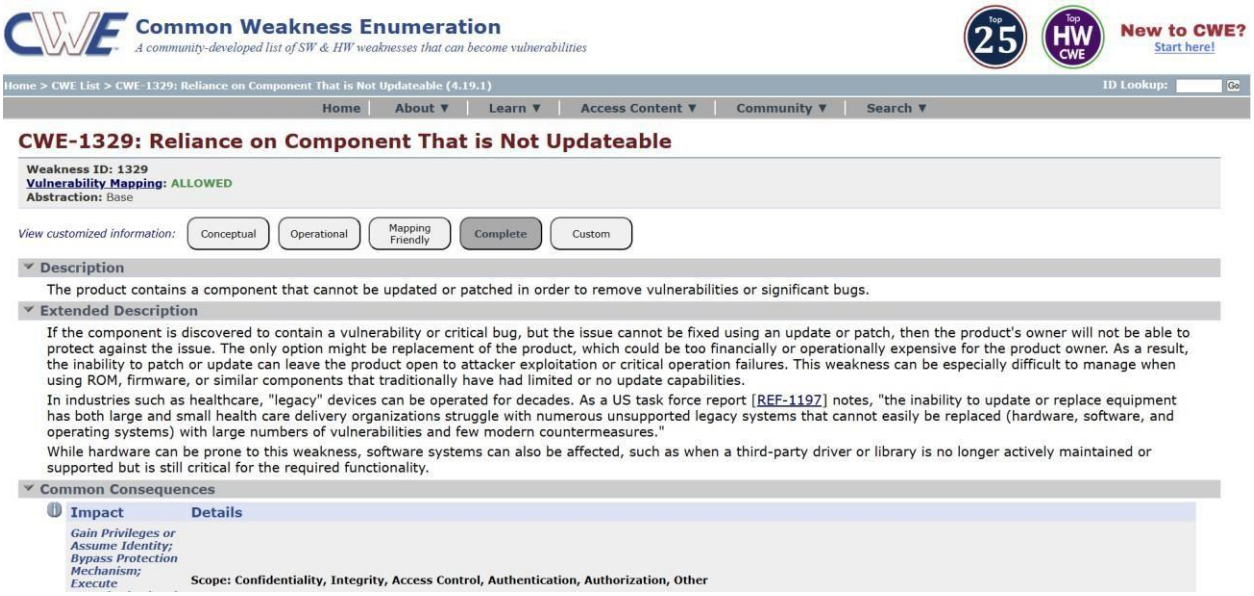
Business Impact:

Reliance on components that cannot be updated can expose organizations to serious security threats. Attackers often target outdated software components because their vulnerabilities are

publicly known and easier to exploit. If these components cannot be updated or patched, the system may remain permanently vulnerable to cyberattacks.

This situation can lead to unauthorized access to the application, leakage of sensitive information, or disruption of critical business services. Attackers may also exploit vulnerable components to inject malicious code or gain control over system functions.

For organizations, such security incidents can result in financial losses, reputational damage, and loss of customer trust. In some cases, exposure of sensitive data may also lead to legal consequences and regulatory penalties. Maintaining outdated or unsupported components can therefore significantly increase operational risk and reduce the overall reliability of the application.



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CWE-1329: Reliance on Component That is Not Updateable

Weakness ID: 1329
Vulnerability Mapping: ALLOWED
Abstraction: Base

View customized information: Conceptual | Operational | Mapping Friendly | Complete | Custom

Description

The product contains a component that cannot be updated or patched in order to remove vulnerabilities or significant bugs.

Extended Description

If the component is discovered to contain a vulnerability or critical bug, but the issue cannot be fixed using an update or patch, then the product's owner will not be able to protect against the issue. The only option might be replacement of the product, which could be too financially or operationally expensive for the product owner. As a result, the inability to patch or update can leave the product open to attacker exploitation or critical operation failures. This weakness can be especially difficult to manage when using ROM, firmware, or similar components that traditionally have had limited or no update capabilities.

In industries such as healthcare, "legacy" devices can be operated for decades. As a US task force report [REF-1197] notes, "the inability to update or replace equipment has both large and small health care delivery organizations struggle with numerous unsupported legacy systems that cannot easily be replaced (hardware, software, and operating systems) with large numbers of vulnerabilities and few modern countermeasures."

While hardware can be prone to this weakness, software systems can also be affected, such as when a third-party driver or library is no longer actively maintained or supported but is still critical for the required functionality.

Common Consequences

Impact	Details
Gain Privileges or Assume Identity; Bypass Protection Mechanism; Execute	Scope: Confidentiality, Integrity, Access Control, Authentication, Authorization, Other

Recommendation / Mitigation:

• Organizations should regularly review and maintain an inventory of all third-party components, libraries, and dependencies used within their applications.

• Software components should be updated to the latest secure versions whenever updates or security patches are released by the developers.

• Avoid using outdated or unsupported libraries that no longer receive maintenance or security updates.

• Implement dependency management tools to monitor and track vulnerable components within the software supply chain.

• Conduct regular security assessments and vulnerability scanning to detect outdated or insecure software components.

• Use trusted and verified sources when integrating third-party software to reduce the risk of introducing vulnerable or malicious components into the system.

1. **Vulnerability Name:** Key Exchange without Entity Authentication

CWE/CVE: CWE-322

OWASP/SANS Category: OWASP Top 10 – Cryptographic Failures

Description:

This vulnerability occurs when a cryptographic key exchange process is performed without properly verifying the identity of the communicating parties. In secure communication systems, key exchange mechanisms are used to establish encryption keys between two entities so that sensitive data can be transmitted securely. However, if the system does not authenticate the entities involved in the key exchange, attackers may intercept the communication and impersonate one of the parties.

Without entity authentication, an attacker can perform a man-in-the-middle attack and establish separate encryption keys with both parties while secretly monitoring or modifying the transmitted data. As a result, confidential information may be exposed or altered without the knowledge of the legitimate users. This weakness undermines the security of the cryptographic process and reduces the effectiveness of encryption mechanisms.

Vulnerability Path:

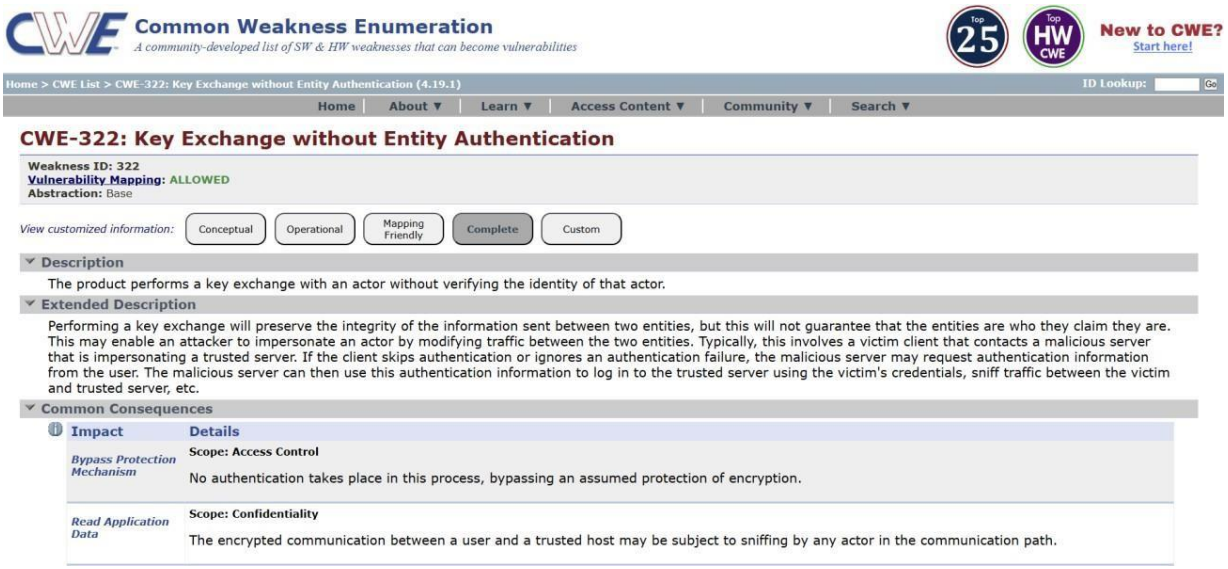
http://example.com/api/secure-communication

Business Impact:

Cryptographic failures caused by improper key exchange mechanisms can lead to serious security risks for organizations. If attackers intercept the communication during the key exchange process, they may gain access to sensitive data such as login credentials, personal information, or financial transactions.

This vulnerability may allow attackers to eavesdrop on confidential communications, modify transmitted data, or impersonate legitimate users within the system. Such attacks can compromise the integrity and confidentiality of sensitive information and may disrupt secure business operations.

For organizations, these security breaches can result in financial losses, reputational damage, and loss of customer trust. Exposure of confidential data may also lead to regulatory penalties and legal consequences, especially when organizations fail to protect sensitive user information properly.



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CWE-322: Key Exchange without Entity Authentication

Weakness ID: 322
Vulnerability Mapping: ALLOWED
Abstraction: Base

View customized information: Conceptual | Operational | Mapping Friendly | Complete | Custom

Description

The product performs a key exchange with an actor without verifying the identity of that actor.

Extended Description

Performing a key exchange will preserve the integrity of the information sent between two entities, but this will not guarantee that the entities are who they claim they are. This may enable an attacker to impersonate an actor by modifying traffic between the two entities. Typically, this involves a victim client that contacts a malicious server that is impersonating a trusted server. If the client skips authentication or ignores an authentication failure, the malicious server may request authentication information from the user. The malicious server can then use this authentication information to log in to the trusted server using the victim's credentials, sniff traffic between the victim and trusted server, etc.

Common Consequences

Impact	Details
Bypass Protection Mechanism	**Scope: Access Control** No authentication takes place in this process, bypassing an assumed protection of encryption.
Read Application Data	**Scope: Confidentiality** The encrypted communication between a user and a trusted host may be subject to sniffing by any actor in the communication path.

Recommendation / Mitigation:

• Implement secure key exchange protocols that include proper entity authentication to verify the identity of both communicating parties.

• Use strong and well-established cryptographic protocols such as TLS with certificate-based authentication to protect communication channels.

• Ensure that digital certificates are properly validated before establishing secure connections between systems.

• Avoid using weak or outdated cryptographic algorithms that may expose the system to security risks.

• Regularly review and update cryptographic implementations to follow modern security standards and best practices.

• Conduct regular security testing and code reviews to identify weaknesses in cryptographic processes and improve overall system security.

2. **Vulnerability Name:** Improper Neutralization of HTTP Headers for Scripting Syntax

CWE/CVE: CWE-644

OWASP/SANS Category: OWASP Top 10 – Injection

Description

This vulnerability occurs when a web application fails to properly validate or sanitize HTTP header inputs before processing them. HTTP headers such as User-Agent, Referer, or Host may contain malicious scripting syntax inserted by an attacker. If the application processes these headers without neutralizing the malicious content, the injected script may be executed within the application environment or reflected in responses sent to other users. Attackers can exploit this weakness to inject malicious scripts or commands through manipulated HTTP headers. This may lead to unauthorized actions, execution of malicious code, or manipulation of application

behavior. Improper validation of header inputs therefore creates an opportunity for attackers to exploit injection vulnerabilities and compromise application security.

Vulnerability Path:

http://example.com/page

Malicious HTTP Header

User-Agent: <script>alert('attack')</script>

Business Impact:

Injection vulnerabilities related to improper neutralization of HTTP headers can create serious security risks for web applications. When attackers are able to insert malicious scripting syntax into HTTP header fields, the application may process this data without proper validation or filtering. As a result, malicious scripts may be executed within the application environment or reflected in responses sent to users.

Attackers can exploit this vulnerability to manipulate application behavior, steal session information, or gain unauthorized access to sensitive user data. In some cases, injected scripts may allow attackers to hijack user sessions, perform unauthorized actions on behalf of users, or modify the content displayed by the application.

For organizations, such attacks may lead to data breaches, service disruption, and compromise of confidential information. Exposure of sensitive data can damage the organization's reputation and reduce customer trust. In addition, security incidents caused by injection vulnerabilities may result in financial losses and potential legal or regulatory consequences.

Common Weakness Enumeration
A community-developed list of SW & HW weaknesses that can become vulnerabilities

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CWE-644: Improper Neutralization of HTTP Headers for Scripting Syntax

Weakness ID: 644
Vulnerability Mapping: ALLOWED
Abstraction: Variant

View customized information: Conceptual | Operational | Mapping Friendly | Complete | Custom

Description

The product does not neutralize or incorrectly neutralizes web scripting syntax in HTTP headers that can be used by web browser components that can process raw headers, such as Flash.

Extended Description

An attacker may be able to conduct cross-site scripting and other attacks against users who have these components enabled.

If a product does not neutralize user controlled data being placed in the header of an HTTP response coming from the server, the header may contain a script that will get executed in the client's browser context, potentially resulting in a cross site scripting vulnerability or possibly an HTTP response splitting attack. It is important to carefully control data that is being placed both in HTTP response header and in the HTTP response body to ensure that no scripting syntax is present, taking various encodings into account.

Common Consequences

Impact	Details
Execute Unauthorized Code or Commands	**Scope: Integrity, Confidentiality, Availability** Run arbitrary code.
Read Application Data	**Scope: Confidentiality** Attackers may be able to obtain sensitive information.

Recommendation / Mitigation:

• All HTTP header inputs should be properly validated and sanitized before being processed by the application.

• Applications should implement strict input validation mechanisms to detect and block malicious scripting syntax in request headers.

• Output encoding should be applied when displaying header values in web pages to prevent execution of injected scripts.

• Security libraries and frameworks should be used to automatically handle input validation and reduce the risk of injection vulnerabilities.

• Web application firewalls should be deployed to detect and block malicious requests containing header-based injection attempts.

• Regular security testing and vulnerability scanning should be conducted to identify and fix injection weaknesses within the application.

3. **Vulnerability Name:** Trust Boundary Violation

CWE/CVE: CWE-501

OWASP/SANS Category: OWASP Top 10 – Insecure Design

Description:

This vulnerability occurs when an application incorrectly trusts data that crosses a security boundary without performing proper validation or verification. In secure system design, a trust boundary separates components with different levels of trust, such as user input, external systems, or internal services. When data moves from a less trusted area to a more trusted component without adequate validation, it can lead to a trust boundary violation. Attackers can exploit this weakness by sending malicious or manipulated data to the application, which may then be processed as if it were trustworthy. As a result, the system may perform unauthorized actions, expose sensitive information, or allow attackers to bypass security controls.

Vulnerability Path:

http://example.com/api/userdata

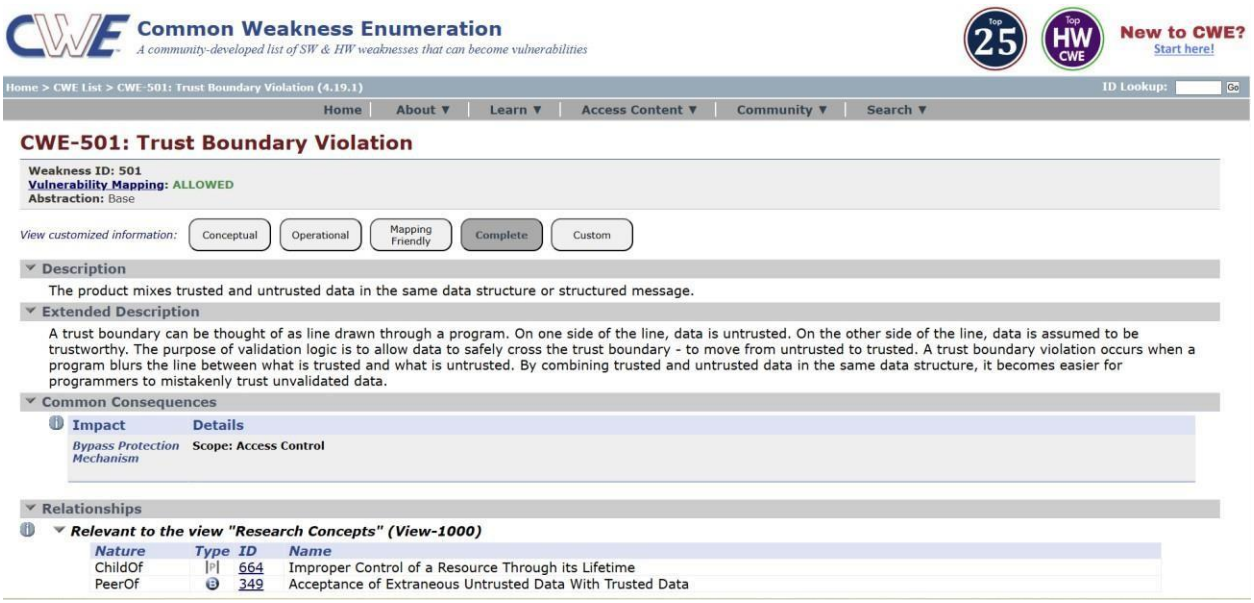
Business Impact:

Trust boundary violations can lead to serious security risks within an application because untrusted data may be processed without proper validation. Attackers can exploit this weakness to manipulate system behavior, bypass security checks, or gain unauthorized access to sensitive resources. When the application incorrectly trusts external inputs or third-party systems, malicious data may enter critical components of the system.

This vulnerability can allow attackers to modify application logic, perform unauthorized transactions, or access confidential information stored within the system. In some cases, attackers may escalate privileges or interfere with internal system operations.

For organizations, such security weaknesses can result in data breaches, service disruption, and financial losses. Exposure of sensitive information may damage the organization's reputation and

reduce customer confidence in the system. In addition, organizations may face legal or regulatory consequences if they fail to properly protect sensitive user or business data.



Recommendation / Mitigation:

• Applications should clearly define trust boundaries and ensure that all data crossing these boundaries is properly validated and sanitized.

• Strong input validation mechanisms should be implemented to verify data received from users, external systems, or third-party services.

• Security controls should be applied at every layer of the application to prevent untrusted data from affecting critical system components.

• Developers should follow secure design principles and perform threat modeling during the system design phase.

• Access control mechanisms should be enforced to ensure that only authorized entities can interact with sensitive system components.

• Regular security testing and design reviews should be conducted to identify and fix weaknesses related to trust boundary violations.

4. **Vulnerability Name:** Authentication Bypass by Alternate Name

CWE/CVE: CWE-289

OWASP/SANS Category: OWASP Top 10 – Authentication Failures

Description:

This vulnerability occurs when an application allows authentication to be bypassed by using an alternate name or identifier that the system incorrectly accepts as valid. In many applications, authentication mechanisms verify user identity using usernames, email addresses, account IDs, or other identifiers. If the system fails to properly validate these identifiers, an attacker may exploit this weakness by providing an alternate or manipulated name to bypass authentication controls. As a result, the attacker may gain unauthorized access to the system without providing valid credentials. This weakness typically occurs when authentication checks rely on inconsistent identity verification or when multiple identifiers are treated as equivalent without proper validation.

Vulnerability Path:

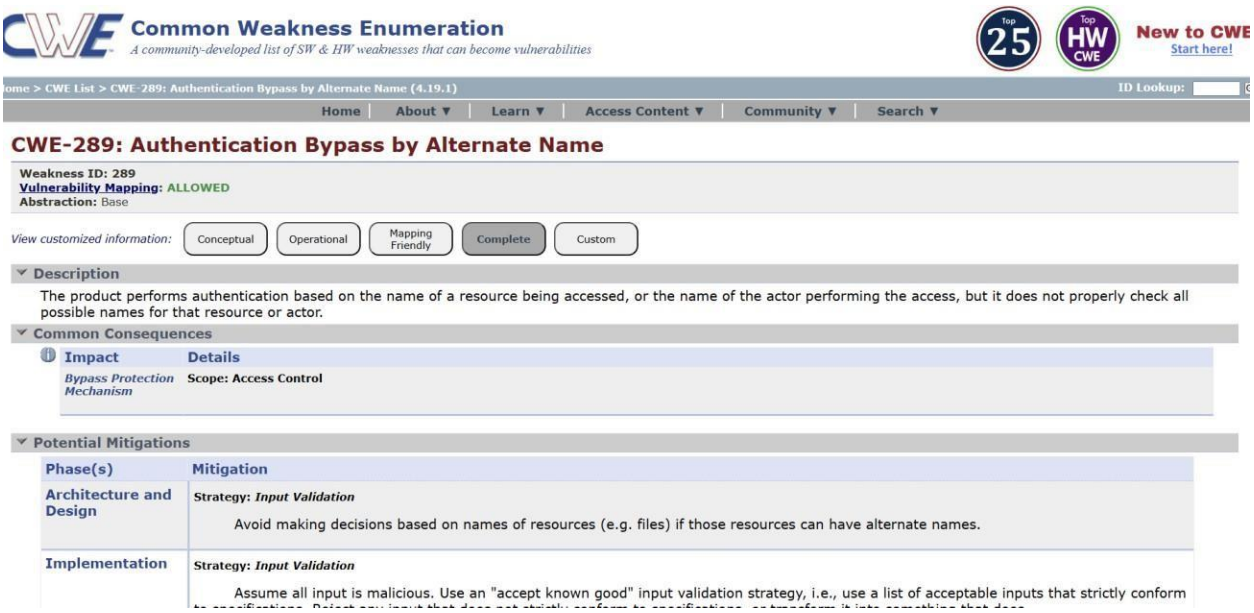
http://example.com/login?username=admin

Business Impact:

Authentication bypass vulnerabilities can lead to serious security risks because attackers may gain unauthorized access to user accounts or administrative functions. If an attacker successfully bypasses authentication controls, they may access sensitive information, modify application data, or perform actions on behalf of legitimate users.

This vulnerability may allow attackers to impersonate other users or administrators within the system. As a result, confidential information such as personal data, financial records, or internal business information may be exposed or manipulated. Attackers may also disrupt system operations or misuse system resources.

For organizations, such security incidents can result in data breaches, financial losses, and operational disruption. Unauthorized access to sensitive accounts may damage the organization's reputation and reduce customer trust. In addition, organizations may face legal or regulatory consequences if they fail to properly secure authentication mechanisms.



Common Weakness Enumeration

A community-developed list of SW & HW weaknesses that can become vulnerabilities

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CWE-289: Authentication Bypass by Alternate Name

Weakness ID: 289
Vulnerability Mapping: ALLOWED
Abstraction: Base

View customized information: Conceptual | Operational | Mapping Friendly | Complete | Custom

Description

The product performs authentication based on the name of a resource being accessed, or the name of the actor performing the access, but it does not properly check all possible names for that resource or actor.

Common Consequences

Impact	Details
Bypass Protection Mechanism	Scope: Access Control

Potential Mitigations

Phase(s)	Mitigation
Architecture and Design	Strategy: Input Validation Avoid making decisions based on names of resources (e.g. files) if those resources can have alternate names.
Implementation	Strategy: Input Validation Assume all input is malicious. Use an "accept known good" input validation strategy, i.e., use a list of acceptable inputs that strictly conform

Recommendation / Mitigation:

• Applications should implement strong authentication mechanisms to ensure that all user identities are properly verified before granting access.

• All user identifiers such as usernames, email addresses, or account IDs should be strictly validated to prevent authentication bypass attempts.

• Multi-factor authentication should be implemented to provide an additional layer of security for user accounts.

• Access control mechanisms should ensure that authentication checks cannot be bypassed using alternate identifiers.

• Proper logging and monitoring should be enabled to detect suspicious login attempts or authentication anomalies.

• Regular security testing and code reviews should be conducted to identify and fix weaknesses in authentication mechanisms.

5. **Vulnerability Name:** Insufficient Verification of Data Authenticity

CWE/CVE: CWE-345

OWASP/SANS Category: OWASP Top 10 – Software or Data Integrity Failures

Description:
This vulnerability occurs when an application fails to properly verify the authenticity or integrity of data before processing it. In many systems, data may come from external sources such as user inputs, third-party services, software updates, or external files.

If the application does not verify whether the data is genuine and has not been tampered with, attackers may inject or modify data before it is processed by the system. As a result, the application may trust and execute malicious or altered data. This weakness can allow attackers to manipulate application behavior, distribute malicious software updates, or alter critical system data without detection.

Vulnerability Path:

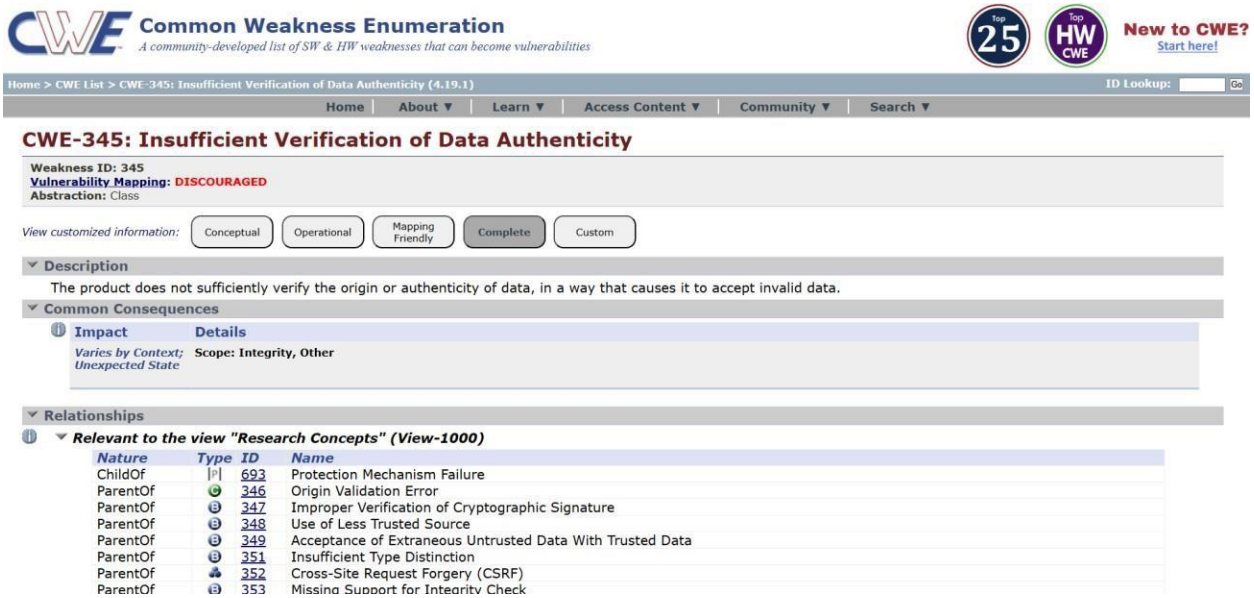
http://example.com/update/download

Business Impact:

Insufficient verification of data authenticity can expose organizations to serious security risks because the system may process untrusted or modified data as legitimate. Attackers may exploit this vulnerability to inject malicious files, modify application updates, or manipulate critical business data.

If malicious data is accepted by the application, it may lead to unauthorized system changes, execution of harmful code, or corruption of important information. Attackers may also exploit this weakness to distribute malware through software updates or compromise system functionality.

For organizations, such attacks can result in data breaches, operational disruption, and financial losses. The integrity of business data may be compromised, which can affect decision-making processes and system reliability. In addition, exposure of sensitive information or distribution of compromised software can damage the organization's reputation and lead to legal or regulatory consequences.



Recommendation / Mitigation:

• Applications should implement strong integrity verification mechanisms to ensure that all incoming data is authentic and has not been modified.

• Digital signatures or cryptographic hash functions should be used to verify the authenticity and integrity of software updates and external data.

• Data received from external sources should always be validated and verified before being processed by the system.

• Secure communication channels should be used to prevent unauthorized modification of data during transmission.

• Access controls should be implemented to restrict who can modify or upload critical data within the system.

• Regular security monitoring and integrity checks should be conducted to detect any unauthorized data changes.

6. **Vulnerability Name:** Information Loss or Omission

CWE/CVE: CWE-221

OWASP/SANS Category: OWASP Top 10 – Security Logging and Monitoring Failures

Description:
This vulnerability occurs when an application fails to properly record important security events or omits critical information from system logs. Logging and monitoring mechanisms are essential for detecting suspicious activities, identifying security incidents, and supporting forensic investigations. If important events such as login attempts, system errors, data access, or configuration changes are not recorded correctly, security teams may not be able to detect attacks or investigate incidents effectively. Information loss or omission in logs can occur due to improper logging configuration, incomplete log generation, or failure to capture essential system activities. As a result, attackers may perform malicious actions within the system without being detected.

Vulnerability Path:

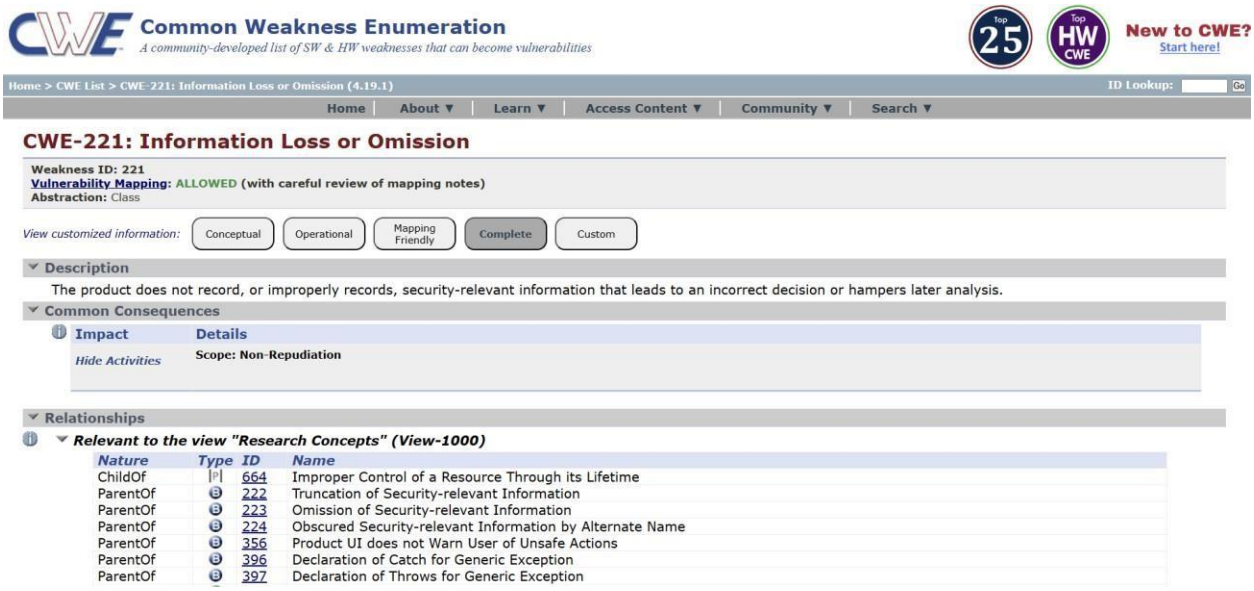
http://example.com/login

Business Impact:

Security logging and monitoring failures can significantly reduce an organization's ability to detect and respond to cyberattacks. When important system activities are not properly logged, attackers may exploit vulnerabilities, access sensitive data, or perform unauthorized actions without leaving sufficient evidence in the system logs.

This vulnerability can make it difficult for administrators to identify the source of security incidents, track malicious activities, or determine how an attack occurred. As a result, organizations may experience delayed responses to security breaches and increased damage to their systems and data.

For organizations, insufficient logging and monitoring can lead to data breaches, operational disruptions, and financial losses. In addition, failure to maintain proper security logs may violate regulatory or compliance requirements. This can damage the organization's reputation and reduce customer trust in the system's ability to protect sensitive information.



Recommendation / Mitigation:

• Applications should implement comprehensive logging mechanisms to record all important security events and system activities.

• Logs should include detailed information such as timestamps, user identifiers, IP addresses, and the type of activity performed.

• Security monitoring tools should be used to analyze logs and generate alerts for suspicious or abnormal behavior.

• Log files should be securely stored and protected from unauthorized modification or deletion.

• Regular log reviews and audits should be conducted to detect potential security incidents and investigate abnormal activities.

• Organizations should implement centralized logging systems to ensure consistent monitoring across all applications and infrastructure components.

7. **Vulnerability Name:** NULL Pointer Dereference

CWE/CVE: CWE-476

OWASP/SANS Category: Mishandling of Exceptional Conditions

Description

This vulnerability occurs when a program attempts to use a pointer that has a NULL value without performing proper checks. In many programming languages, a NULL pointer represents the absence of a valid memory reference. If the application tries to access or manipulate data through a NULL pointer, the program may crash or behave unexpectedly. This situation often occurs when error conditions are not properly handled or when developers assume that a pointer always contains a valid memory address. Attackers may intentionally trigger such conditions by providing unexpected input or manipulating application behavior. As a result, the system may fail to handle exceptional situations correctly, leading to application crashes or denial of service conditions.

Vulnerability Path:

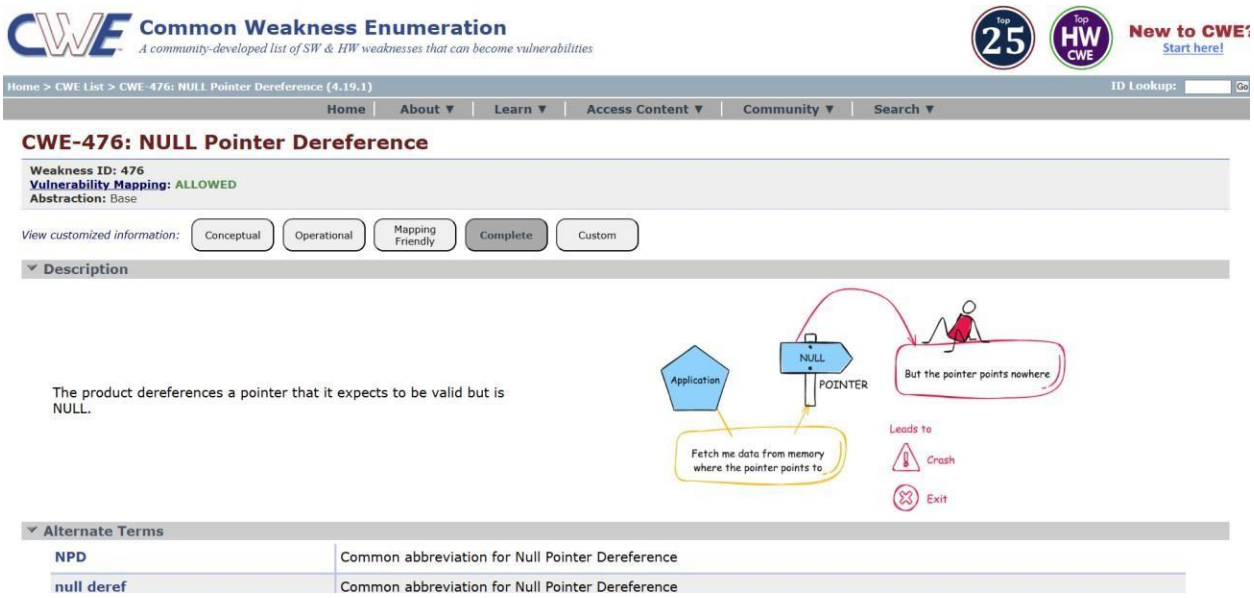
http://example.com/api/getUser?id=

Business Impact:

NULL pointer dereference vulnerabilities can cause applications to crash or stop functioning when the system attempts to access invalid memory references. Attackers may exploit this weakness by sending specially crafted requests that trigger the error condition, resulting in denial of service. When the application fails repeatedly, legitimate users may be unable to access the system or complete important operations.

Such failures can disrupt business operations, reduce system availability, and negatively affect user experience. In some cases, repeated crashes may also expose debugging information or internal system details that attackers could use to identify additional vulnerabilities.

For organizations, these issues can lead to service interruptions, operational downtime, and potential financial losses. Frequent system failures may damage the organization's reputation and reduce customer confidence in the reliability of the application.



Recommendation / Mitigation:

• Developers should always perform proper NULL checks before accessing or using pointers within the application.

• Error handling mechanisms should be implemented to safely manage unexpected or invalid conditions in the program.

• Input validation should be applied to ensure that the application does not process invalid or incomplete data.

• Secure coding practices should be followed to ensure that pointers are properly initialized before use.

• Regular code reviews and static analysis tools should be used to detect potential NULL pointer dereference issues during development.

• Comprehensive testing should be conducted to identify exceptional conditions and ensure that the application handles them safely without crashing.

Stage 2

Overview :

Nessus is a widely used vulnerability assessment tool designed to identify security weaknesses in computer systems, networks, and applications. It is developed by **Tenable** and is commonly used by security professionals, network administrators, and organizations to improve system security. The main purpose of Nessus is to scan systems and detect vulnerabilities that could potentially be exploited by attackers.

Nessus works by performing automated vulnerability scans on target systems. During the scanning process, the tool sends various network requests to the target machine and analyzes the responses received. By examining these responses, Nessus can collect information about the operating system, running services, open ports, installed applications, and system configurations. Based on this information, the scanner determines whether the system contains any known vulnerabilities.

One of the key features of Nessus is its **plugin-based architecture**. Plugins are small programs that are designed to detect specific security issues. Each plugin checks for a particular vulnerability such as outdated software versions, missing security patches, weak passwords, insecure configurations, or exposed network services. Nessus contains thousands of plugins, and these plugins are continuously updated to detect newly discovered vulnerabilities.

The scanning process usually begins with **host discovery**, where the tool identifies active systems on the network. After discovering the target systems, Nessus performs **port scanning** to determine which ports and services are open. Once the services are identified, the scanner runs appropriate plugins to check those services for potential vulnerabilities. This automated process helps detect security weaknesses efficiently without manually checking each system.

After the scanning process is completed, Nessus generates a detailed vulnerability report. The report lists all detected vulnerabilities along with important information such as the vulnerability name, severity level, description, possible impact, and recommended solutions.

Vulnerabilities are usually categorized into different severity levels including **Critical, High, Medium, Low, and Informational**. This classification helps administrators prioritize which issues should be fixed first.

Overall, Nessus plays an important role in cybersecurity by helping organizations identify weaknesses in their systems before attackers can exploit them. Regular vulnerability scanning using Nessus helps improve system security, reduce risks, and maintain a safer network environment.

Target website : https://demo.testfire.net/
Target ip address: 65.61.137.117

List of vulnerability

s.no	Vulnerability name	Severity	plugins
1	SSL/TLS Diffie-Hellman Modulus <= 1024 Bits (Logjam)	LOW	83875
2	ICMP Timestamp Request Remote Date Disclosure	LOW	10114
3	HyperText Transfer Protocol (HTTP) Information	INFO	24260
4	HSTS Missing From HTTPS Server	INFO	84502

5	TLS Version 1.0 Protocol Detection	MEDIUM	104743
6	TLS Version 1.1 Deprecated Protocol	MEDIUM	157288
7	SSL Cipher Suites Supported	INFO	21643
8	SSL Root Certification Authority Certificate Information	INFO	94761
9	Nessus SYN scanner	INFO	11219
10	OS Identification	INFO	11936

REPORT

1. **Vulnerability Name:** SSL/TLS Diffie-Hellman Modulus <= 1024 Bits (Logjam)

Severity : LOW

Plugin: 83875

Port : 443

Description: The remote host allows SSL/TLS connections with one or more Diffie-Hellman moduli less than or equal to 1024 bits. Through cryptanalysis, a third party may be able to find the shared secret in a short amount of time (depending on modulus size and attacker resources). This may allow an attacker to recover the plaintext or potentially violate the integrity of connections.

Solution: Reconfigure the service to use a unique Diffie-Hellman moduli of 2048 bits or greater.

Business Impact: The presence of weak **SSL/TLS Diffie-Hellman keys (≤1024 bits)** can reduce the strength of encrypted communications between the server and its users. Attackers with sufficient computing resources may potentially break the encryption and gain access to sensitive information transmitted over the network. This could lead to **data exposure, loss of confidentiality, and unauthorized interception of communications**.

For organizations, this vulnerability may result in **reputational damage, reduced customer trust, and possible non-compliance with modern security standards and regulatory requirements**. Although the severity is classified as **low**, leaving weak cryptographic configurations in place can still weaken the overall security posture and make the s ystem more vulnerable to advanced cryptographic attacks.

2. Vulnerability Name: ICMP Timestamp Request Remote Date Disclosure

Severity : LOW

Plugin: 10114

Port : 0

Description: The remote host answers to an ICMP timestamp request. This allows an attacker to know the date that is set on the targeted machine, which may assist an unauthenticated, remote attacker in defeating time-based authentication protocols. Timestamps returned from machines running Windows Vista / 7 / 2008 / 2008 R2 are deliberately incorrect, but usually within 1000 seconds of the actual system time.

solution:

Filter out the ICMP timestamp requests (13), and the outgoing ICMP timestamp replies (14).

Business Impact: Allowing **ICMP timestamp requests** can expose the system's internal date and time information to external users. Although the vulnerability is considered **low severity**, attackers can use this information to gather details about the target system during the reconnaissance phase of an attack.

By knowing the system time, an attacker may be able to **analyze server behavior, assist in bypassing certain time-based authentication mechanisms, or coordinate attacks more effectively**. This information disclosure may also help attackers combine this vulnerability with other weaknesses to perform more advanced attacks.

For organizations, this could lead to **increased risk of targeted attacks, reduced system privacy, and potential exploitation when combined with other vulnerabilities**, ultimately affecting the overall security posture of the network.

3. Vulnerability Name: HyperText Transfer Protocol (HTTP) Information

severity : INFO

Plugin:24260

Port :80

Description: This test gives some information about the remote HTTP protocol - the version used, whether HTTP Keep-Alive is enabled, etc... This test is informational only and does not denote any security problem.

solution: -

Business Impact: This finding is **informational** and does not represent a direct security vulnerability. However, it reveals details about the **HTTP service configuration**, such as the protocol version and connection settings. Attackers may use this information during the **reconnaissance phase** to better understand the target system and identify potential weaknesses.

Although the risk is minimal, exposing unnecessary system information can help attackers **plan more targeted attacks if other vulnerabilities exist**. For organizations, this may slightly increase the risk of **information disclosure and system profiling**, which could support further exploitation when combined with other security issues.

4. Vulnerability Name: HSTS Missing From HTTPS Server

severity : INFO

Plugin: 84502

Port :443

Description: The remote HTTPS server is not enforcing HTTP Strict Transport Security (HSTS). HSTS is an optional response header that can be configured on the server to instruct the browser to only communicate via HTTPS. The lack of HSTS allows downgrade attacks, SSL- stripping man-in-the-middle attacks, and weakens cookie-hijacking protections.

solution: Configure the remote web server to use HSTS.

Business Impact: The absence of **HTTP Strict Transport Security (HSTS)** means that users may still access the website through **unencrypted HTTP connections** before being redirected to HTTPS. This creates an opportunity for attackers to perform **man-in-the-middle attacks, SSL stripping, or session hijacking**, where sensitive information such as login credentials or session cookies could be intercepted.

Although this finding is categorized as **informational**, it weakens the protection of secure communications between users and the website. For organizations, this may lead to **data interception risks, reduced user trust, and potential exposure of sensitive information** if attackers exploit the lack of enforced HTTPS communication.

5. Vulnerability Name: TLS Version 1.0 Protocol Detection

severity : MEDIUM

Plugin: 104743

Port :443

Description: The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 has a number of cryptographic design flaws. Modern implementations of TLS 1.0 mitigate these problems, but newer versions of TLS like 1.2 and 1.3 are designed against these flaws and should be used whenever possible.

As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher will no longer function properly with major web browsers and major vendors.

PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2018, except for POS POI terminals (and the SSL/TLS termination points to which they connect) that can be verified as not being susceptible to any known exploits.

solution: Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

Business Impact: The use of **TLS 1.0** exposes the system to several known cryptographic weaknesses and outdated security mechanisms. Attackers may exploit these weaknesses to perform **man-in-the-middle attacks, data interception, or decryption of sensitive communications** between users and the server.

Since **TLS 1.0 is deprecated and no longer supported by many modern browsers and security standards**, its presence can also lead to **compliance issues with regulations such as PCI DSS**. This may result in **security audit failures, potential penalties, and operational risks** for the organization.

If exploited, this vulnerability could compromise the **confidentiality and integrity of sensitive data**, damage the organization's reputation, and reduce user trust in the security of the system. Enabling **TLS 1.2 or TLS 1.3** significantly improves encryption strength and helps maintain secure communications.

6. Vulnerability Name: TLS Version 1.1 Deprecated Protocol

severity : MEDIUM

Plugin:157288

Port :443

Description: The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 lacks support for current and recommended cipher suites. Ciphers that support encryption before MAC computation, and authenticated encryption modes such as GCM cannot be used with TLS 1.1

As of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher will no longer function properly with major web browsers and major vendors.

solution: Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

Business Impact: The use of **TLS 1.1**, which is a deprecated protocol, weakens the security of encrypted communications between the server and its users. Since TLS 1.1 does not support modern and stronger cipher suites, it may expose the system to **known cryptographic weaknesses and potential data interception attacks**.

In addition, many **modern browsers and security standards no longer support TLS 1.1**, which may lead to **compatibility issues, service disruption, and compliance failures** with industry security regulations. If attackers exploit these weaknesses, sensitive data transmitted over the network could be compromised.

For organizations, this can result in **loss of data confidentiality, reduced customer trust, and potential regulatory non-compliance**, affecting the overall security and reliability of the system. Enabling **TLS 1.2 or TLS 1.3** ensures stronger encryption and better protection for data in transit.

7. Vulnerability Name: SSL Cipher Suites Supported

severity : INFO

Plugin:21643

Port :443

Description: This plugin detects which SSL ciphers are supported by the remote service for encrypting communications.

solution: -

Business Impact: This finding is **informational** and does not directly indicate a security vulnerability. It simply identifies the **SSL/TLS cipher suites supported by the server** for encrypting communications.

However, exposing the list of supported cipher suites may help attackers **analyze the encryption configuration of the server**. If weak or outdated cipher suites are enabled, attackers could attempt to exploit them to **reduce encryption strength or intercept secure communications**.

For organizations, this information could assist attackers in **identifying potential weaknesses in the encryption setup**, which may increase the risk of future attacks if stronger and recommended cipher suites are not enforced.

8. Vulnerability Name: SSL Root Certification Authority Certificate Information

severity :INFO

Plugin:94761

Port :443

Description: The remote service uses an SSL certificate chain that contains a self-signed root Certification Authority certificate at the top of the chain.

solution: Ensure that use of this root Certification Authority certificate complies with your organization's acceptable use and security policies.

Business Impact: This finding is **informational** and does not indicate a direct security vulnerability. It only provides details about the **root Certification Authority (CA) certificate** used in the SSL/TLS certificate chain.

However, revealing certificate information may help attackers **analyze the certificate structure and trust chain** of the server. If the certificate is not properly managed or trusted, it could lead to **trust issues with clients or browsers**, potentially affecting secure communication.

For organizations, improper management of root CA certificates may lead to **security policy violations, trust warnings in browsers, or reduced confidence in the system's security** , which could impact user trust and the reliability of encrypted communications.

9. Vulnerability Name: Nessus SYN scanner

severity : INFO

Plugin:11219

Port :80

Description: This plugin is a SYN 'half-open' port scanner. It shall be reasonably quick even against a firewalled target.

Note that SYN scans are less intrusive than TCP (full connect) scans against broken services, but they might cause problems for less robust firewalls and also leave unclosed connections on the remote target, if the network is loaded.

solution: Protect your target with an IP filter.

Business Impact: This finding is **informational** and does not indicate a direct vulnerability in the system. It simply shows that a **SYN scan was used to identify open ports and services** on the target host during the security assessment.

However, the ability to perform such scans indicates that the system may allow **network reconnaissance activities**, which attackers can use to discover open ports and running services. This information can help attackers **map the network and identify potential entry points for future attacks**.

For organizations, this may increase the risk of **unauthorized network scanning, service discovery, and targeted attacks** if proper network filtering or intrusion detection mechanisms are not in place. Implementing **IP filtering, firewalls, and monitoring systems** can help reduce this risk and improve overall network security.

10. Vulnerability Name: OS Identification

severity : INFO

Plugin:11936

Port :N/A

Description: Using a combination of remote probes (e.g., TCP/IP, SMB, HTTP, NTP, SNMP, etc.), it is possible to guess the name of the remote operating system in use. It is also possible sometimes to guess the version of the operating system.

solution: -

Business Impact: This finding is **informational** and does not represent a direct security vulnerability. It indicates that the operating system of the target host can be **identified or estimated through network responses and service fingerprints**.

However, if attackers are able to determine the **operating system and its version**, they can use this information to **search for known vulnerabilities specific to that OS**. This can make it easier for attackers to plan and launch **targeted attacks or exploits** against the system.

For organizations, this may increase the risk of **system profiling and targeted exploitation**, especially if the identified operating system has **unpatched vulnerabilities or outdated components**. Limiting unnecessary information disclosure and keeping systems updated can help reduce this risk.

Stage 3

Report

Tittle : REPLACE AND MODERNIZE LEGACY VPN ARCHITECTURES WITH A CLOUD-NATIVE ZSCALER ZERO TRUST SOLUTION TO REDUCE ATTACK SURFACE AND LATERAL MOVEMENT.

Security Operations Center (SOC)

A Security Operations Center (SOC) is a centralized unit that monitors, detects, and responds to cybersecurity threats in an organization. In this project, SOC helps monitor user access when applications are accessed through the Zscaler Zero Trust platform instead of traditional VPNs. By integrating authentication with Microsoft Azure Active Directory, SOC teams can monitor login activities and detect suspicious behavior to prevent unauthorized access.

SOC – cycle

The SOC cycle includes monitoring, detection, investigation, response, and recovery. Security tools collect logs from systems and applications. These logs are analyzed to detect threats. When suspicious activity is found, the SOC team investigates the issue and takes action to stop the threat and restore normal operations.

Security Information and Event Management (SIEM)

SIEM is used to collect and analyze security logs from different systems in a network. It helps organizations detect suspicious activities and potential cyber threats. In this project, SIEM monitors authentication and access logs generated when users connect through Zscaler and Azure Active Directory.

SIEM Cycle

The SIEM cycle includes log collection, data analysis, correlation, and alert generation. Logs are collected from multiple systems and analyzed for abnormal activities.

If suspicious behavior is detected, the SIEM system generates alerts so that security teams can investigate and respond quickly.

Malware Information Sharing Platform (MISP)

MISP is an open-source threat intelligence platform that helps organizations share information about cyber threats. It provides details about malware, vulnerabilities, and malicious activities, helping security teams detect and prevent attacks earlier.

Your college network information

The college network consists of computers, servers, routers, and switches connected through a local network. Students and staff use this network to access learning resources and institutional data. To protect sensitive information, strong security monitoring and authentication systems are necessary.

How you think you deploy soc in your college

SOC can be deployed in the college network to monitor security events and detect threats. Security logs from servers, authentication systems, and network devices can be analyzed using SIEM tools. By using Zscaler with Azure Active Directory, secure authentication and continuous monitoring can be implemented to protect the network.

Threat intelligence

Threat intelligence is the process of collecting and analyzing information about cyber threats. It helps organizations understand attack patterns and improve their security defenses.

Incident response

Incident response is the process of detecting, investigating, and resolving cybersecurity incidents. When a threat is detected, the security team identifies the problem, contains the attack, removes the threat, and restores the system.

Qradar & understanding about tool

IBM QRadar is a SIEM platform used for security monitoring and threat detection. It collects security logs from multiple systems and analyzes them to identify suspicious activities. The dashboard helps security analysts monitor events and investigate incidents efficiently.

Conclusion

Stage 1

Stage 1 focused on understanding how Zscaler can be integrated with identity providers such as Microsoft Azure Active Directory. It helped in learning about secure authentication, Zero Trust access, and identifying possible security vulnerabilities in cloud environments.

Stage 2

Stage 2 helped in understanding vulnerability scanning and security analysis using Nessus reports to identify and fix potential security weaknesses.

Stage 3

Stage 3 focused on advanced cybersecurity concepts such as SOC, SIEM, threat intelligence, and incident response. These technologies help organizations monitor network activity, detect cyber threats, and respond quickly to security incidents, improving overall network security.

Future Scope :

Stage 1

In the future, the integration of Zscaler with identity providers can be expanded with stronger authentication methods and additional security mechanisms. Automated security assessment tools can also be used to detect vulnerabilities more efficiently.

Stage 2

Future improvements in vulnerability testing can include advanced scanning techniques and automated security tools such as Nessus to identify and fix security weaknesses more efficiently.

Stage 3

Future improvements may include the use of Artificial Intelligence and Machine Learning for faster threat detection and automated incident response. Expanding SOC monitoring and integrating more security tools will further strengthen cybersecurity.

Topics explored :

- Cloud Security
- Identity Provider Integration
- Security Operations Center
- SIEM Monitoring
- Threat Intelligence
- Incident Response

Tools explored :

- Zscaler
- Microsoft Azure Active Directory
- IBM QRadar
- MISP Threat Intelligence Platform

THE END